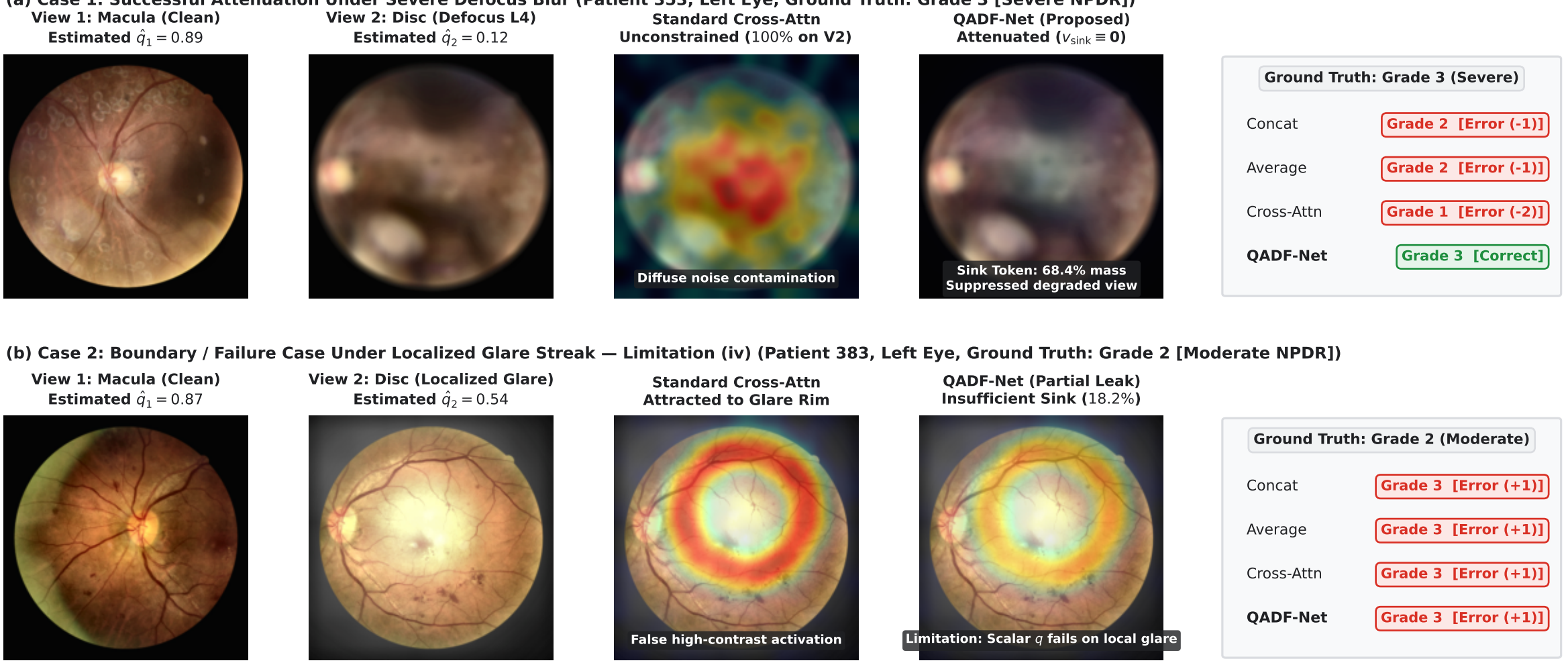

(a) Case 1: Successful Attenuation Under Severe Defocus Blur (Patient 353, Left Eye, Ground Truth: Grade 3 [Severe NPDR])
View 1: Macula (Clean)
Estimated $\hat{q}_1 = 0.89$
View 2: Disc (Defocus L4)
Estimated $\hat{q}_2 = 0.12$
Standard Cross-Attn
Unconstrained (100% on V2)
Diffuse noise contamination
QADF-Net (Proposed)
Attenuated ($v_{sink} \equiv 0$)
Sink Token: 68.4% mass
Suppressed degraded view
Ground Truth: Grade 3 (Severe)
Concat
Grade 2 [Error (-1)]
Average
Grade 2 [Error (-1)]
Cross-Attn
Grade 1 [Error (-2)]
QADF-Net
Grade 3 [Correct]
(b) Case 2: Boundary / Failure Case Under Localized Glare Streak — Limitation (iv) (Patient 383, Left Eye, Ground Truth: Grade 2 [Moderate NPDR])
View 1: Macula (Clean)
Estimated $\hat{q}_1 = 0.87$
View 2: Disc (Localized Glare)
Estimated $\hat{q}_2 = 0.54$
Standard Cross-Attn
Attracted to Glare Rim
False high-contrast activation
QADF-Net (Partial Leak)
Insufficient Sink (18.2%)
Limitation: Scalar q fails on local glare
Ground Truth: Grade 2 (Moderate)
Concat
Grade 3 [Error (+1)]
Average
Grade 3 [Error (+1)]
Cross-Attn
Grade 3 [Error (+1)]
QADF-Net
Grade 3 [Error (+1)]